

Effect of Pore Confinement of Ionic Liquids on Solute Diffusion within Mesoporous Silica Microparticles

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Cite This: *J. Phys. Chem. B* 2024, 128, 3046–3060



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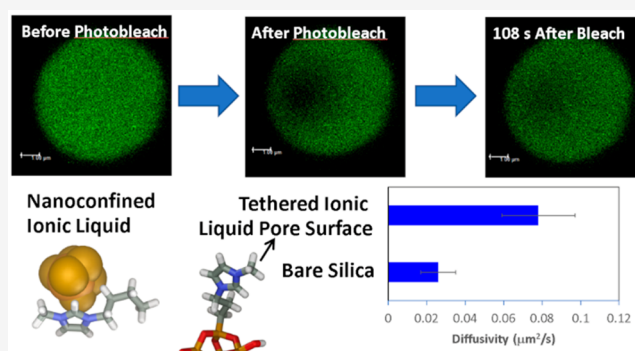


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ABSTRACT: The transport properties of the ionic liquid (IL) 1-butyl-3-methylimidazolium hexafluorophosphate ([BMIM][PF₆]) confined within silica microparticles with well-ordered, accessible mesopores (5.4 or 9 nm diameter) were investigated. [BMIM][PF₆] confinement was confirmed by using differential scanning calorimetry (DSC) and Fourier transform infrared (FTIR) spectroscopy. The transport properties of the confined IL were studied using the neutral and cationic fluorescent probes 4-(dicyanomethylene)-2-methyl-6-(4-dimethylaminostyryl)-4H-pyran (DCM) and rhodamine 6G, respectively, through fluorescence recovery after photobleaching (FRAP) in confocal microscopy. The diffusivity of DCM in 9 nm pores is $0.026 \pm 0.0091 \mu\text{m}^2/\text{s}$, which is 2 orders of magnitude less than in the bulk ionic liquid. The pore size did not affect the diffusivity of DCM in unmodified silica nanopores. The diffusivity of the cationic probe is reduced by 63% relative to that of the neutral probe. Diffusivity is increased with water content, where equilibrium hydration of the system leads to a 37% increase in DCM diffusivity. The most dramatic impact on diffusivity was caused by tethering an IL-like methylimidazolium chloride group to the pores, which increased the pore hydrophobicity and resulted in 3-fold higher diffusivity of DCM compared to bare silica pores. Subsequent exchange of the chloride anion from the tethering group with PF₆[−] decreased the diffusivity to half that of bare silica. The diffusion of probe molecules is affected most strongly by the pore wall effects on probe interactions rather than by the pore size itself, which suggests that understanding pore wall diffusion is critical to the design of nanoconfined ILs for separations, catalysis, and energy storage.



1. INTRODUCTION

Ionic liquids (ILs) are salts that are in a liquid state near ambient conditions, typically with melting points below 100 °C,¹ where the combination of cation or anions can be changed to modify their solvent properties² and allow them to function as low vapor pressure “designer solvents”.³ ILs exist as a liquid at low temperatures because the component ions are large in size and have conformational flexibility, resulting in the liquid state being thermodynamically favorable.⁴ Applications of ILs include the removal of toxic organics⁵ or metals⁶ from water, the extraction of organic materials from waste streams,⁵ as solvents in catalysis,⁷ for CO₂ capture,⁸ as electrolytes in capacitors for energy storage,⁹ and many others.¹⁰ Several factors limit the use of ILs as bulk phase solvents: their cost, their viscosity and corresponding low mass transfer rate,^{11,12} and their potential toxicity,¹³ emphasizing the importance of recovering ILs even in dilute solutions. Immobilization of ILs on surfaces or their confinement in a porous matrix with submicrometer pathlength takes advantage of their favorable solvent properties while mitigating the limitations of transport and solvent recovery in bulk phases.^{8,14–16}

Confinement of ILs in nanoscale pores alters their physical properties compared to those of the bulk phase. For example,

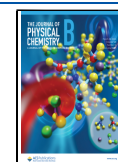
shifts in the melting temperature (or other characteristic temperatures) of a confined IL have been observed as a function of pore wall chemistry, ionic liquid components, and the method of confinement (e.g., forming an ionogel or confining the IL after synthesis of the substrate).¹⁷ Zuo et al. found by differential scanning calorimetry (DSC) that the glass transition temperature of 1-butyl-3-methylimidazolium tetrafluoroborate and 1-butyl-3-octylimidazolium tetrafluoroborate confined between alumina surfaces increased as the confinement length decreased due to interactions between the BF₄[−] anion and the positively charged alumina surface.¹⁸ Other techniques to probe the interactions between ILs and pore walls in confined systems include solid-state NMR¹⁹ and FTIR.¹⁷ Molecular dynamics has been extensively used to study the layering of the ionic liquid cation with the pore walls

Received: February 15, 2024

Revised: February 28, 2024

Accepted: February 29, 2024

Published: March 13, 2024



through charge interactions,^{14,20,21} from which changes in the physical properties are inferred. ILs confined in mesoporous silica have been investigated as catalysts for asymmetric catalysis, Friedel–Crafts isopropylation, and trimerization reactions.^{22–24} Confined ILs are also promising candidates for gas adsorption. Their high surface area, tunable properties, and high adsorption capabilities due to high affinity toward CO₂ make them ideal for carbon capture in gas separations.²⁵ Additionally, confinement reduces the amount of IL necessary to be used in practical applications, making them less cost prohibitive and improving safety by not allowing them to freely flow into waste streams or other solvents. Thus, in many applications, confinement of ILs into mesoporous materials makes them more attractive than bulk solvents.

In addition to physically loading the ILs on solid support or in pores, mesoporous silica has been modified by covalent attachment of IL-like moieties (such as 1-propyl-3-methylimidazolium chloride)²⁶ to the accessible silanol groups on the pore walls. Subsequently, an additional (free) IL can be incorporated in the tethered pores. This method of confinement prevents leaching of the tethered IL and offers more stability of the supported IL.²⁷ Tethering an IL-like molecule to the surface changes the orientation and interactions of free IL with the surface, which is expected to affect the corresponding solvent properties in the pores. For instance, [BMIM][PF₆] and its analogues interact through H-bonding and electrostatic attraction of the cationic imidazolium ring to the surface of silica.¹ In the system of 1-octylimidazolium bis(trifluoromethylsulfonyl)imide confined within trioctylsilyl-grafted silica particles, the alkyl tail of the IL is oriented toward the functionalized surface of the silica if the pores are <5 nm.²⁸ The surface can also be modified after tethering through ion exchange, with a goal of changing the counterion of the tethered moiety and thus the IL itself. For example, the water contact angle of silica with tethered IL 1-alkyl-3-(3-triethoxysilylpropyl)imidazolium chloride on the surface increased from 24° to 42° after anion exchange from [Cl][−] to [PF₆][−], indicating that the surface became more hydrophobic.²⁹ The choice of anion is important depending on the intended application of the material. For carbon capture applications, the CO₂ solubility in [BMIM][PF₆] is greater than that in [BMIM][Cl],³⁰ so having [PF₆][−] as the anion would be a better choice for the tethered species.

Mesoporous silica is frequently used as the solid substrate for the confinement of ILs because robust synthesis procedures lead to uniform, accessible, and tunable nanopores (2–20 nm) that are readily functionalized with various groups including amines or thiols.^{31–33} Specifically, mesoporous silica provides high pore surface area and pore volume for IL loading.^{15,34–36} Mesoporous silica can be synthesized as thin films, as nano- or microparticles, or directly on membrane supports,^{37–39} which can be broadly translated to applications of mesoporous silica in drug delivery, biosensing, separations, and catalysis. The sol–gel synthesis of ordered mesoporous silica used in this study uses self-assembled surfactants or block copolymers to form templates for the pores, followed by silica condensation with coassembled micelles, and removal of the surfactant or block copolymer template to result in ordered, well-defined pores.⁴⁰

This study uses micrometer-sized SBA-15 type mesoporous silica particles, which have well-ordered hexagonally arranged pores with a tunable pore size (5–15 nm). In addition, the micrometer-sized particles are sufficiently large to be imaged

with confocal microscopy. An important benefit of supported or confined ILs as separation or reaction solvents relative to bulk ILs is the reduced transport pathlength through these viscous fluids. Traceable fluorescent probe molecules have been used to measure the translational diffusion in bulk ILs. Specifically, confocal microscopy has been used to study diffusion rates in bulk ionic liquids⁴¹ through visualization of dye transport measured as fluorescence recovery after photobleaching (FRAP). FRAP takes advantage of the photobleaching of dyes excited by a laser in a defined area of the sample and the recovery of fluorescence over time as unbleached probe molecules diffuse back into the bleached spot. Other groups have used fluorescence correlation spectroscopy (FCS),^{42,43} a technique based on autocorrelation analysis of fluctuations of fluorescence intensity within a constant volume, to study the diffusivity of dyes in bulk ionic liquid systems. While the two methods provide similar information in principle, they differ in that FCS probes smaller volume and time scales (<1 s) while FRAP probes larger volume and time scales (>10 s and lower diffusivities than FCS).⁴⁴ Diffusion in bulk ILs is fast enough for FCS to be applied. Werner et al. used FCS to study the diffusivity within neat [BMIM][PF₆] using three differently charged dyes—rhodamine 6G (cationic), DCM (neutral), and fluorescein (anionic)—and found that the anionic dye had the fastest diffusivity, followed by the neutral dye, with the cationic dye being the slowest. Additionally, when water was added to the system, the diffusivity increased by approximately 65–70% for all dyes.⁴³ Sasmal et al. measured the diffusivity of three separate dyes within neat 3-pentyl-1-methylimidazolium bromide, [PMIM][Br], and [PMIM][BF₄] using FCS and found that the dye charge and hydrophobicity played a significant role in the diffusivity of the probe from either attractive forces between the probes and IL or the reorientation of the probe within the IL.⁴² FCS shows that changes to the probe or the environment in bulk solvents led to significant changes in transport.

The transport properties in IL-containing systems are further affected by nanoconfinement. Many molecular dynamics studies have been performed of ILs near surfaces and confined in pores.⁴⁵ Specifically for [BMIM][PF₆] confined in 7.2 nm nanoslits with amorphous silica walls, Pal et al. reported a significant reduction in diffusivity at 300 K, where a strongly ordered first layer of ions at the pore wall had roughly 2 orders of magnitude less mobility than at the center of the pore.⁴⁶ In their case, the silica was modeled as a fully protonated, neutral surface with PF₆[−] bound in the first layer by hydrogen bonding;⁴⁷ in real systems, deprotonation of silica is likely to lead to stronger charge interactions with [BMIM]⁺ and possibly a stronger impact on diffusion. Similar findings of slowed diffusion near solid surfaces were found by MD for silica^{48,49} and other materials such as carbon.^{50,51} One study of graphene oxide nanopores suggested the possibility of enhanced diffusion due to the creation of free volume by movement of IL ions to the pore wall.⁵² Nuclear magnetic resonance experiments have shown diffusion and relaxation dynamics consistent with slower dynamics due to nanoconfinement,^{53,54} for example, with 2 orders of magnitude slower diffusion of both [EMIM]⁺ cations and bis(fluorosulfonyl)imide anions.⁵⁵ Electrochemical measurements also show reductions in conductivity for nanoconfined ILs.^{56–58} Small ions used by these techniques such as Li⁺ and H⁺ are able to freely diffuse within nanoconfined ILs, and self-diffusion is not significantly

limited within mesopores,^{28,53} but larger molecules such as dyes can interact with the IL and change its diffusive properties.^{43,59–61} While there has been several studies using fluorescence to measure diffusion of organic probes in bulk phase ILs and self-diffusion using NMR,^{53,55} there are still opportunities to study the diffusion of solutes within confined ILs. Additionally, there are no studies of the effect of pore surface functionalization on the diffusion of probe molecules though confined ILs except for in mesoporous thin films using electrochemistry.⁶²

In this work, we explore the diffusion of dye molecules within a confined IL to probe effects of confinement using confocal microscopy. Confocal microscopy is employed to directly probe the effect of confinement on transport of fluorescent probe molecules in physically confined and tethered [BMIM][PF₆] within synthesized mesoporous silica microparticles with 5.4 and 9 nm diameter pores. The present work demonstrates that the FRAP technique is a suitable method to characterize transport of organic molecules through confined ILs. FRAP measures the transport properties of solutes dissolved within the IL rather than the self-diffusion coefficients of the IL components that can be obtained by NMR. FRAP is also advantageous over electrochemical impedance spectroscopy because the recovery curve can be directly modeled using elementary diffusion equations rather than being inferred from resistances. It also does not require an external driving source for the probe molecule. [BMIM][PF₆] is a commonly studied hydrophobic ionic liquid that has been researched extensively as a model IL due to its relatively low viscosity and low melting point. A hydrophobic IL was required because the particles were suspended in water during the confocal experiments to maintain similar refractive indices between the microscope and the sample. The bare particles were characterized with nitrogen porosimetry to determine the pore diameter, and FTIR and DSC were used to confirm confinement of the IL. The effect of pore size was studied by changing the average pore diameter of the silica through the hydrothermal aging temperature. Neutral (DCM) and cationic (R6G) probes were used to understand the effect of the dye charge on transport. The effect of water uptake was studied by immersion of the particles with confined IL in water for 24 h because most ILs have at least a small amount of water in them and readily pick up water from the atmosphere or contact of hydrophobic ILs with aqueous media. The walls of the silica particles were functionalized with an IL-like molecule, 3-methyl-1-[3-(trimethoxysilyl)propyl]-1*H*-imidazol-3-ium chloride ([TMS-MIM][Cl]), to examine how tethering (which is a strategy to control IL leaching) might affect the diffusion of the probe molecules. The results below show that the transport through the confined IL on mesoporous silica is primarily dominated by surface and electrostatic interactions. The insensitivity of transport in supported ILs to pore diameter and water uptake (shown below) also means that they are flexible and can be used in situations where water may be present.

2. METHODS

2.1. Materials. The ionic liquid 1-butyl-3-methylimidazolium hexafluorophosphate (>97%) ([BMIM][PF₆]) was purchased from Sigma-Aldrich. 200 proof anhydrous ethanol was purchased from Koptec (Deacon Laboratories). For the synthesis of microparticles, Pluronic 123 was purchased from BASF, tetraethyl orthosilicate (99%) was

purchased from Alfa Aesar, hydrochloric acid (6 N) was purchased from VWR Chemicals BDH, and cetyltrimethylammonium bromide (CTAB) (99.8%) was purchased from MP Biomedicals. The dyes used were 4-(dicyanomethylene)-2-methyl-6-(4-dimethylaminostyryl)-4*H*-pyran (DCM) (>95%) from TCI America and rhodamine 6G (R6G) (>99%) from Acros Organics. Chloroform was purchased from Macron Fine Chemicals. All materials were used as received without further purification.

2.2. Silica Microparticle Synthesis. The method for synthesis of micrometer-sized mesoporous silica particles with controlled pore size via hydrothermal treatment was adapted from Gartmann's modified procedure⁶³ from the work of Katiyar⁶⁴ and previously used in our lab.⁶⁵ Initially, a few grams of P123 in a beaker was melted in an oven at 80 °C, of which 3.10 g was poured into a round-bottom flask. 0.465 g of CTAB was dissolved in 20 mL of DI water, stirred for 10 min, and added to the melted P123. This solution was stirred continuously at room temperature throughout the rest of the procedure. 7.8 mL of 200 proof ethanol and 49.5 mL of 1.5 M HCl were added to the solution and stirred until a clear, homogeneous solution formed, which took about 45 min. Particle formation was initiated by adding 10 mL of TEOS dropwise to the solution and stirring the mixture for 2 h. The solution was then poured into a Parr 4748A Teflon-lined bomb (200 mL capacity) at room temperature and then placed into a preheated oven and aged for 72 h. The pore size of the particles was modified by adjusting the thermal aging temperature⁶³ where large pore size particles (9 nm) were aged at 125 °C and smaller pore size particles (5.4 nm) were aged at 65 °C. After three full days, the particles were removed from the bomb and mixed with a high-speed mixer to homogenize the solution, then filtered, and rinsed with DI water. The particles were then placed in a single-walled Whatman cellulose extraction thimble and underwent Soxhlet extraction with 200 mL of ethanol for 24 h. Afterward, the particles were dried under vacuum at 80 °C overnight and characterized.

2.3. [TMS-MIM][Cl] Synthesis and Tethering and Ion Exchange. The tethering IL 3-methyl-1-[3-(trimethoxysilyl)propyl]-1*H*-imidazol-3-ium chloride ([TMS-MIM][Cl]) was synthesized by first making a mixture of *N*-methylimidazole (2.00 mL, 25 mmol) and (3-chloropropyl)trimethoxysilane (4.59 mL, 25 mmol) in toluene (40 mL) and refluxing the mixture for 36 h. After cooling to room temperature, the supernatant solution was removed with a pipet. The remaining oil was washed with 10 mL of diethyl ether three times to remove any remaining reactants and then dried under reduced pressure at room temperature. The product was collected as a colorless oil (6.80 g, 97%). ¹H and ¹³C NMR spectra of the prepared IL-like organosilane confirmed the identity of the material, as reported in the literature.^{66,67} Toluene and diethyl ether were dried and distilled by standard procedures prior to use.⁶⁸

The procedure for tethering the ionic liquid presented by Gruttadauria et al.²² was adapted to functionalize the silica pore walls. Initially, 100 mg of silica microparticles (BJH adsorption surface area of 445.7 m²/g) with a pore diameter of 9 nm was vacuum-dried at 140 °C overnight. 0.20 g of the tethering ionic liquid, [TMS-MIM][Cl] (0.73 mmol), was dissolved into 15 mL of chloroform in a nitrogen-filled glovebag, and the silica microparticles were added to the solution. The mixture was refluxed for 26 h at 65 °C and then

cooled to room temperature. The microparticles were filtered from the solution, washed with 50 mL of acetonitrile and 50 mL of diethyl ether, and then dried overnight to produce a white solid, the tethered 9 nm silica particles. Nitrogen adsorption was used to verify that the pore diameter and volume had decreased due to functionalization of the pore walls.

A portion of the tethered silica particles underwent further modification to exchange the chloride anion to the desired $[\text{PF}_6]^-$ based on the procedure by Vangeli et al.^{27,29} Briefly, ion exchange began by stirring 25 mg of tethered particles in 1 mL of a 67.5 mM solution of NaPF_6 in water. After 24 h, the particle solution was centrifuged, and the spent NaPF_6 solution was decanted and mixed with 3 drops of 1% AgNO_3 in DI water. If a cloudy appearance was present (indicative of chloride ions), the particles were dispersed into fresh NaPF_6 solution and left to stir for an additional 24 h. The ion exchange procedure was repeated until no chloride ions were detected by AgNO_3 . Afterward, the particles were rinsed with methanol and refluxed with 5 mL of methanol at 70 °C for 30 min, filtered, and dried.

2.4. Silica Microparticle Characterization. Nitrogen adsorption was used to characterize the pore diameter distributions and total specific surface area of the microparticles before and after functionalization and IL loading using a TriStar 3000 automatic sorption analyzer (Micromeritics Instrument Corp., Norcross, GA). The instrument uses volumetric adsorption of nitrogen at 77 K to allow calculation of the specific surface area and pore diameter based on the Brunauer, Emmett, and Teller (BET) isotherm and adsorption branch of the Barrett, Joyner, and Halenda (BJH) method, respectively.⁶⁵ Prior to the characterization, samples were degassed at 140 °C for a minimum of 4 h under flowing nitrogen.

Small-angle X-ray scattering (SAXS) data were collected on the unloaded (no IL present) particles to further characterize the pore-to-pore distance. The silica particles were mounted in the powder sample holder of a Xeuss 2.0 SAXS laboratory beamline (Xenocs Inc., Holyoke, MA) equipped with a 30 W microfocus Cu K α source ($\lambda = 1.54 \text{ \AA}$) and a Pilatus3 200K detector. Scattering data were collected for 10 min in transmission SAXS mode. The sample–detector distance was set at 2.3 m calibrated using silver behenate. The locations of the (100) peaks were determined by azimuthal integration of the 2D patterns.

A Nicolet Nexus 470 FTIR instrument (Thermo Fisher Scientific, Waltham, MA) was used to collect IR spectra of the samples from 400 to 4000 cm^{-1} at room temperature to confirm the confinement of the IL and tethering of $[\text{TMS-MIM}][\text{Cl}]$ to the mesoporous silica. Spectra were taken with a resolution of 1.928 cm^{-1} by using at least 300 scans. The sample chamber is continuously purged with dry, low CO_2 air to prevent interference from water vapor and CO_2 gas. Samples were pressed into KBr pellets at about 1 wt %, and data were collected approximately 2 min after the samples were placed into the sample chamber to purge ambient air from the chamber. The FTIR spectrum of neat $[\text{BMIM}][\text{PF}_6]$ liquid was collected using a multireflection attenuated total reflectance (ATR) accessory (PIKE Technologies, Fitchburg, WI) equipped with a ZnSe trough.

2.5. Loading of Silica Microparticles with ILs. Ionic liquids were loaded into mesoporous silica using solvent deposition. The volume of IL that we used to load into the

nanoparticles was determined from the pore volume measured from nitrogen adsorption. In a representative experiment, particles with 9 nm diameter pores (e.g., 9 nm particles) with a total pore volume of 1.51 cm^3/g were vacuum-dried in an oven at 140 °C overnight. Approximately 10 mg of the 9 nm particles, 15.1 μL of IL, and 151 μL of ethanol were sonicated in a 2 mL centrifuge tube for 90 min. Afterward, the particles were vacuum-dried in an oven at 80 °C to remove the ethanol. The amount of IL was scaled accordingly with the change in the pore volume of the 5.4 nm pore particles and tethered particles. IL without a dye was used in FTIR and DSC measurements for all materials.

For confocal microscopy, fluorescent dye was added to the IL prior to loading into the particles. The target molarity of the dye in the IL was 20 μM so that the dye was easily seen in the microscope but not high enough to cause self-quenching. Using DCM as an example, 6.06 mg of the DCM (0.02 mmol) was dissolved in 10 mL of 200 proof ethanol by sonicating for 10 min to make a solution of 2 mM dye. From this solution, 10.1 μL was mixed with 1 mL of $[\text{BMIM}][\text{PF}_6]$ to make a final solution of 20 μM . The final concentration was the same for both DCM and R6G. After the dye was added, the rest of the confinement procedure was the same as neat IL loading. The effect of water on the system was studied by first confining 10 mg of particles with IL as described above; after the particles were dried, 1 mL of water was added to the centrifuge tube. The particles were left immersed in water for 24 h with no agitation before measurement.

2.6. Differential Scanning Calorimetry (DSC) of IL-Loaded Silica Microparticles. DSC measurements were taken to confirm the confinement of the ionic liquid in 9 nm pore diameter bare and tethered mesoporous silica particles and tethered particles. A shift in the direction or disappearance of the cold-crystallization temperature or melting temperature indicates confinement of an IL.⁶⁹ Samples were prepared by placing 5–15 mg of silica into a hermetically sealed aluminum pan in a nitrogen environment to prevent water uptake during the experiment. A Q200 DSC (TA Instruments, New Castle, DE) was used to measure the DSC curves from the sample with a blank reference pan and with the sample chamber purged with continuous flowing nitrogen. Samples were first cooled to $-90 \text{ }^\circ\text{C}$ at $10 \text{ }^\circ\text{C}/\text{min}$ and held isothermally for 10 min, then heated to $25 \text{ }^\circ\text{C}$ at a rate of $10 \text{ }^\circ\text{C}/\text{min}$, and held isothermally for 10 min. A second cooling step to $-90 \text{ }^\circ\text{C}$ at $10 \text{ }^\circ\text{C}/\text{min}$ was performed, followed by another isothermal step for 10 min. Finally, the sample was heated at a slow rate of $0.5 \text{ }^\circ\text{C}/\text{min}$ to $25 \text{ }^\circ\text{C}$ to complete the experiment.

The shift in melting temperature caused by interfacial free energy in nanoscale liquids is generally described by the Gibbs–Thomson equation.⁷⁰ The relationship for a difference in melting point can be described for liquids confined within cylindrical pores using eq 1:

$$T_{\text{fp}}^0 - T_{\text{fp}} = \frac{\gamma^{\text{SL}} \sigma^{\text{SL}}}{s^{\text{L}} - s^{\text{S}}} \left(\frac{2}{R} \right) = \frac{\gamma^{\text{SL}} T_{\text{fp}}^0 \sigma^{\text{SL}}}{\Delta H_{\text{f}}} \left(\frac{2}{R} \right) \quad (1)$$

where T_{fp}^0 is the equilibrium freezing/melting temperature of a flat interface, T_{fp} is the equilibrium freezing/melting temperature of a curved interface, γ^{SL} is the solid molar volume, s^{L} and s^{S} are the molar entropies of the liquid and solid, respectively, R corresponds to the radius of the pore, and ΔH_{f} is the latent heat of fusion at the flat interface melting temperature.

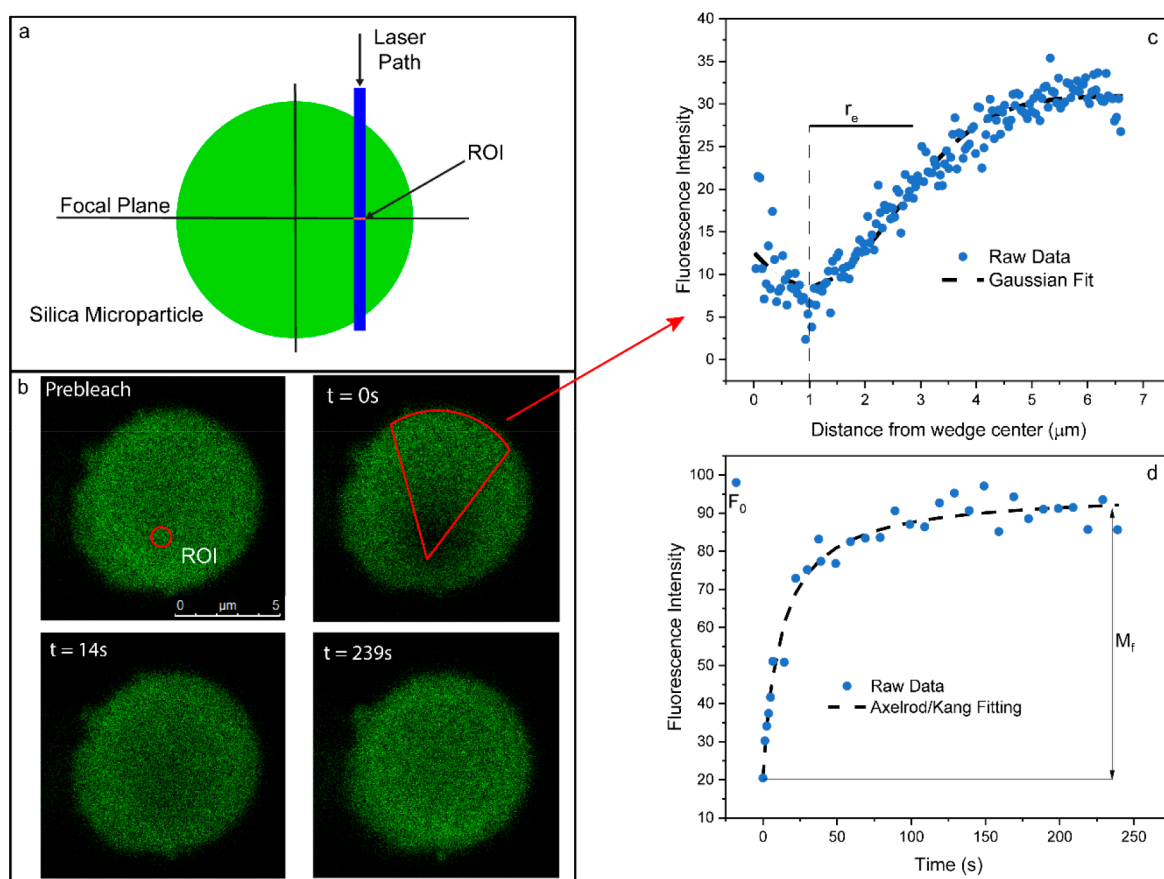


Figure 1. (a) Schematic diagram of the FRAP technique. The location of the measurement (ROI) was taken halfway between the center and the edge of the particle, denoted by the red circle during prebleach measurements in (b) example images taken during a FRAP experiment. The wedge shape at $t = 0$ s, or first postbleach image, was used to fit a Gaussian curve as in panel (c) to find effective radius. The pixels within the ROI were averaged over the course of the experiment and fit with the Axelrod–Kang method, with an example shown in panel (d).

2.7. Confocal Microscopy. Confocal microscopy was performed with a TCS SP5 confocal microscope (Leica Microsystems, Deerfield, IL) equipped with an argon laser using 488 nm as the excitation wavelength for the DCM dye and 538 nm for the rhodamine 6G dye. The temperature of the experiments was room temperature and typically near 22 °C. The objective used in this study was a 63× oil objective with a Type F immersion fluid. Glass slides and coverslips were cleaned using 80% ethanol/water and lens paper to remove smudges and dust. Microparticles were dispersed in DI water just before they were placed into the microscope to maintain a refractive index close to that of the immersion fluid for FRAP measurements. Using a precision pipet, 7 μL of the particle dispersion was placed onto the glass slide, and the coverslip was carefully pressed down. The sample was then placed upside-down in an inverted microscope. A schematic of the photobleach process is shown in Figure 1, along with an example of the data processing. Each FRAP measurement was performed within a focal plane passing through the center (depth) of a microparticle. A circular region of interest (ROI) with a nominal radius of 250 nm (orange outline, Figure 1a) was chosen at the midpoint between the center and the edge of the particle in the focal plane. A 250 nm radius cylindrical laser pattern with the same excitation wavelength as for imaging was used to photobleach the ROI over the course of 6.5 s. Images were collected over a time span of 250 s while the unbleached dye diffused into the bleached area, and the pixel intensity

within the ROI was averaged with the Leica FRAP Wizard software. An example of the raw images before and just after photobleaching ($t = 0$ s), during recovery, and at the end of the experiment is shown in Figure 1b. The total fluorescence intensity in the ROI vs time from the same FRAP experiment is shown as a representative example in Figure 1d.

After data collection, the raw intensity measurements within the ROI were fitted to find the diffusivity. First, the effective radius (r_e) of the bleached spot was determined to account for the diffusion of dye during the photobleaching process by fitting a Gaussian equation (eq 2) to the bleach profile of the first postbleach frame.⁷¹ Kang et al. showed that using the actual effective postbleach radius is essential to accurately estimate diffusivity by FRAP.⁷²

$$F(x) = F_0 + Ae^{-(x-x_c)^2/(r_e^2/\ln(2))} \quad (2)$$

In eq 2, $F(x)$ is the azimuthally averaged pixel fluorescence intensity within a wedge centered near the bleach spot extending radially from 0 to x μm outward toward the unbleached side of the particle (illustrated in Figure 1b at $t = 0$ s), x_c is the center of the bleach spot, and A is the amplitude of the intensity and was obtained with ImageJ⁷³ version 1.53a and the Radial Profile Angle plugin. The effective radius (r_e) is the half-width at half-maximum of the bleach spot. Equation 2 was fit to the fluorescence data by nonlinear regression using

MATLAB version 2020b with F_0 , A , x_c , and r_e as adjustable parameters. A typical fit is illustrated in Figure 1c.

Next, the average intensity within the ROI for each individual FRAP measurement was plotted as a function of time and fitted using the model presented by Kang,⁷⁴ which is a modification of the Axelrod model⁷⁵ for diffusion into a bleached area that initially has a Gaussian intensity profile. The raw intensity values, $F(t)$, were fitted with eq 3:

$$F(t) = F_i \left[1 - \frac{K}{1 + \gamma^2 + \frac{2t}{\tau_D}} \right] M_f + (1 - M_f) F_0 \quad (3)$$

where F_i is the prebleach fluorescence intensity, γ is the ratio of the nominal and effective radius (r_n/r_e), τ_D is the characteristic diffusion time, t is the experiment time in seconds, M_f is the mobile fraction, and F_0 is the initial postbleach fluorescence intensity ($t = 0$ s). The nominal radius r_n is equal to the radius of the ROI or, in other words, the radius of the laser used for FRAP. In this case, $r_n = 250$ nm. K is defined as

$$K = \frac{F_i - F_0}{F_i} (1 + \gamma^2) \quad (4)$$

This model was fit to the $F(t)$ data by nonlinear regression using M_f and τ_D as adjustable parameters with Microsoft Excel's Solver plugin.⁷⁶ After regression, the characteristic diffusion time τ_D was used to calculate the diffusion coefficient using eq 5.

$$D = \frac{r_e^2 + r_n^2}{8\tau_D} \quad (5)$$

The diffusivity values were individually calculated for each measurement and averaged together to yield an average diffusivity value and standard deviation for each sample. The coefficients of determination (R^2) for the fitting of the FRAP data sets are reported in Table S1 of the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Particle Synthesis and Characterization. Spherical silica particles with micrometer-scale particle diameter were chosen for this study because their spherical morphology and scale are appropriate for imaging using confocal microscopy.³⁸ Table 1 lists the pore diameters measured using nitrogen

Table 1. Silica Pore Diameter, Pore Volume, and Surface Area Results from Nitrogen Adsorption and BET Calculations^a

particles	pore diameter (BJH, nm)	pore volume (m ³ /g)	pore surface area (m ² /g)	q peak (Å ⁻¹)	pore-to-pore distance (SAXS, nm)
9 nm particles	9	1.51	445.7	0.0652	11.1
5.4 nm particles	5.4	1.03	653.9	0.091	7.96
tethered 9 nm particles	7.8	0.992	319.4	0.0649	11.2

^aThe tethering on SBA to SBA-T reduces the pore diameter, volume, and surface area significantly due to pore wall functionalization. The q value of the first peak in the SAXS pattern, related to the pore-to-pore distance, is also listed.

adsorption for the base particles used for each experiment, and the pore size distributions of the particles determined by the BJH method are shown in Figure 2. The particles have a

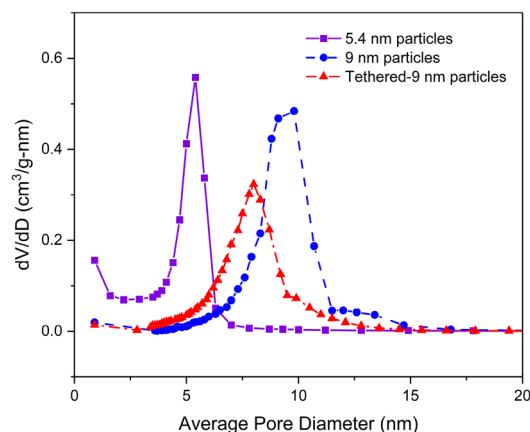


Figure 2. Effect of hydrothermal aging temperature and surface functionalization on the pore diameter distributions. Squares (purple, solid line) represent the bare 5.4 nm mesoporous silica particles hydrothermally aged at 65 °C, circles (blue, dashed line) represent the bare 9 nm mesoporous silica particles aged at 125 °C, and triangles (red, dash-dot) represent the tethered 9 nm silica particles.

uniform mesoporous structure consistent with previous studies.⁶⁵ Increasing the hydrothermal aging temperature from 65 to 125 °C increases the hydrophobic volume of the micelle templates resulting in a larger micelle and pore diameter after synthesis,⁶⁵ leading to mesoporous silica microparticles with 5.4 and 9 nm mesopores, respectively. After the 9 nm porous silica was tethered with [TMS-MIM][Cl] (an IL-like molecule), the pore volume of the particles decreased by 33% from 1.51 to 1.03 cm³/g. The average pore diameter also decreased from ≈ 9 to 8.0 nm. This result is expected because the tethered molecule takes up space on the pore wall surface and leads to a smaller pore volume and pore diameter. After confinement of [BMIM][PF₆] in the pores of the tethered 9 nm silica, nitrogen is unable to adsorb to the pore interiors because the pores are completely filled with IL, resulting in a surface area too small to be accurately measured by nitrogen adsorption (data not shown). Confocal imaging of confined ILs as in Figure 1 demonstrated that the pores are not simply blocked with the IL but are completely full. SAXS was performed on the particles without confined IL, and the results agree with the nitrogen adsorption data. In the Supporting Information, the azimuthally averaged SAXS patterns (Figure S1) have peaks corresponding to a pore-to-pore distance of 11.1, 7.96, and 11.2 nm for 9, 5.4, and tethered 9 nm porous silica, respectively. In combination with the BET analysis, the pore wall thicknesses may be estimated to be 2.1 and 2.6 nm thick in the 9 and 5.4 nm particles, respectively. The pore wall thickness is important because thicker walls are expected to be more stable to structure changes during application.⁷⁷ Because the pore wall thicknesses are similar, the stability of the silica particles is expected to be similar for both pore diameters. The pore diameter in the tethered particles is reduced after functionalization, while pore-to-pore distances remain the same, consistent with the expected effect of tethering.

3.2. Fourier Transform Infrared (FTIR) Spectroscopy. FTIR spectroscopy was used to confirm ionic liquid loading

and tethering in the microparticles. The FTIR spectra of neat and confined $[\text{BMIM}][\text{PF}_6]$ are presented in Figure 3. In the

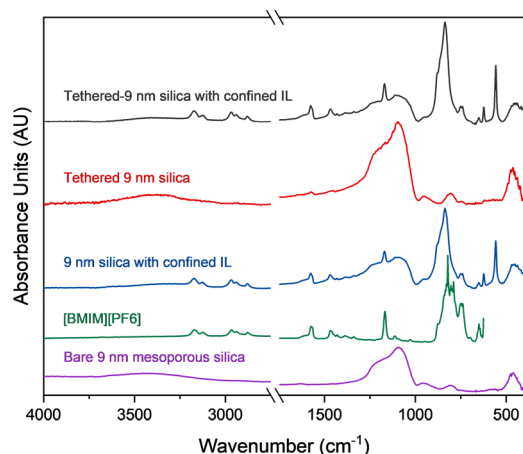


Figure 3. FTIR spectra in the range 4000–400 cm^{-1} for bare 9 nm mesoporous silica particles, pure IL ($[\text{BMIM}][\text{PF}_6]$), confined IL on 9 nm mesoporous silica, tethered-9 nm silica, and tethered-9 nm silica with confined IL.

bare 9 nm silica particles, the peaks at ≈ 1070 (shoulder near 1200), 950, and 800 cm^{-1} are attributed to antisymmetric Si–O–Si stretching, Si–O–H stretching, and symmetric Si–O–Si stretching vibrations, respectively.⁷⁸ In the neat $[\text{BMIM}][\text{PF}_6]$, the peaks between 2800 and 3000 cm^{-1} arise from the ν_s , ν_{as} , and Fermi resonance C–H vibrational modes from the alkyl tail of the IL,⁷⁹ while the features above 3000 cm^{-1} are due to C–H stretching modes in the imidazolium ring.^{1,79} The peaks at 1575 and 1167 cm^{-1} are assigned to stretching modes in the imidazolium ring. The bands between 1500 and 1300 cm^{-1} are attributed to bending modes for CH_2 , CH_3 , and C–CH.⁷⁹ Lastly, the peaks between 750 and 900 cm^{-1} correspond to stretching vibrations of the PF_6^- anion, with smaller peaks at 750 and 650 cm^{-1} being attributed to complex bending and out-of-plane bending, respectively.⁷⁹ The full spectra can be found in Figure S2.

The spectrum of the confined IL within the mesopores of the silica is almost an additive combination of the spectra of bare silica and $[\text{BMIM}][\text{PF}_6]$, but there are slight shifts for peaks associated with the IL. The N=C– stretching, C=C–, and ring stretching peaks shift from 1575 to 1577 cm^{-1} , 1467 to 1469 cm^{-1} , and 1167 to 1170 cm^{-1} , respectively. The higher wavenumber peaks at 3171 and 3124 cm^{-1} shift to 3174 and 3125 cm^{-1} . In the alkyl chain, the peaks shift from 2878, 2938, and 2966 cm^{-1} to 2879, 2941, and 2967 cm^{-1} , respectively. The imidazolium ring peaks have larger shifts to higher energy (higher wavenumber) than the alkyl chain peaks and can be attributed to interactions of the imidazolium ring with the pore walls.^{1,80} Previous FTIR and molecular dynamics studies have demonstrated that the cation from the imidazolium ring can preferentially interact with the silica pore wall.^{1,81} That interaction is found here from the changes to the peaks related to the imidazolium headgroup of the IL. At the same time, the bands associated with PF_6^- shift to higher wavenumbers and may broaden to some extent, reflecting a more heterogeneous solvation environment when confined in silica nanopores. The peaks from organic groups in the tethered but not confined IL are much less pronounced, but new peaks at 1576 and 1461 cm^{-1} are present and correspond

to N=C– and C=C– ring stretching from the IL,¹ providing further evidence that $[\text{TMS-MIM}][\text{Cl}]$ was successfully grafted to the interior pore wall surface. $[\text{BMIM}][\text{PF}_6]$ was confined within the tethered-9 nm silica, and the spectrum is shown in Figure 3. A fully functionalized pore wall would prevent interactions between the IL and the pore wall by eliminating the sites where the interaction could occur. There are no significant differences between the peak locations of the imidazolium groups in the FTIR spectra of confined compared to those of tethered and confined IL. This indicates that some unreacted –OH groups on the pore wall surfaces exist in the tethered-9 nm silica with confined IL for the imidazolium/pore wall interaction to occur.

3.3. Differential Scanning Calorimetry (DSC). The DSC curves used to study the melting point transition temperature of the IL's with and without confinement are presented in Figure 4. The DSC curves for the bare silica

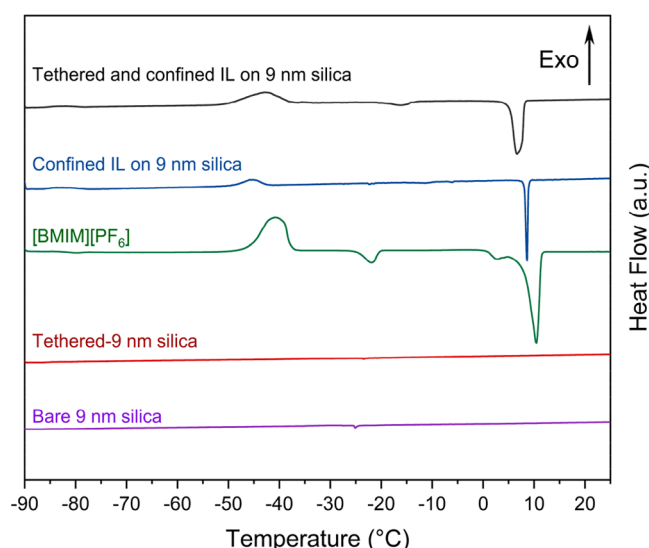


Figure 4. DSC curve of pure $[\text{BMIM}][\text{PF}_6]$, confined IL within 9 nm mesoporous silica particles, and tethered-9 nm particles with confined IL. The heating rate is 0.5 $^{\circ}\text{C}/\text{min}$ from -90 to 25 $^{\circ}\text{C}$ and displays cold crystallization, crystal–crystal phase transition, and melting temperatures. The magnitudes of the curves have been adjusted to fit on the same graph. Bare 9 nm silica particles and tethered silica without confined IL are also included.

substrate and the tethered silica substrate without confined IL demonstrate that the support materials do not contribute to the observed thermal behavior of the confined IL samples. The magnitude of the heat flow values was multiplied by a constant in each curve to fit them onto the same plot. As the temperature is increased, the neat $[\text{BMIM}][\text{PF}_6]$ exhibits a cold crystallization peak at -40 $^{\circ}\text{C}$, two crystal–crystal transition peaks at -21.6 and 2.9 $^{\circ}\text{C}$, and a melting peak at 10.5 $^{\circ}\text{C}$. These are consistent with and close in value to transition temperatures reported in the literature.^{82,83} After confinement, the melting peak of the IL shifts slightly to 8.6 $^{\circ}\text{C}$ and confirms that the IL is confined within the mesoporous silica, as expected, according to the Gibbs–Thomson equation. The melting peak in the confined IL is sharper than the bulk, which could indicate a higher crystallinity or more uniform crystal sizes of the confined IL after cold-crystallization than bulk $[\text{BMIM}][\text{PF}_6]$.^{19,34} Another interesting observation is that the premelting crystal–crystal phase transition processes

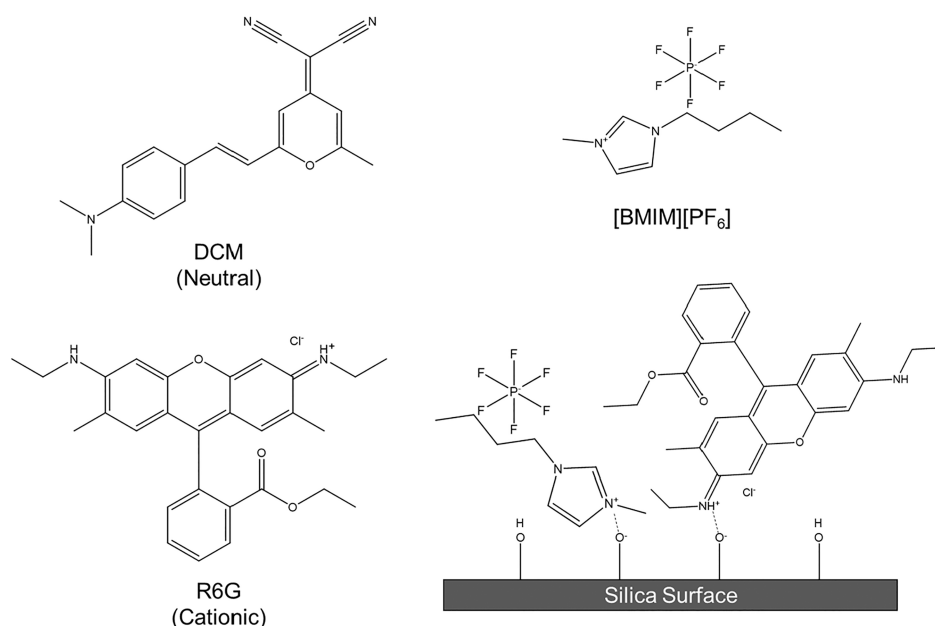


Figure 5. Structures of the ionic liquid [BMIM][PF₆], neutral dye DCM, and cationic dye R6G and illustration of the likely interaction between the IL and R6G with the partially deprotonated silica surface.

at -21.6 and 2.9 °C either no longer occur or their peaks are too small to be seen, perhaps because it is not energetically favorable for the transition to occur under confinement. Another consideration is the orientation of the butyl chain with confinement. The bulk-phase IL contains three conformers in the liquid, supercooled liquid, and glassy states.⁸² The disappearance of the two crystal–crystal phase transition peaks could indicate that only one conformer of the IL is preferred in the confined state. The preference of certain confined IL conformers due to pore wall interactions has also been demonstrated with [BMIM][TFSI] confined within mesoporous silica matrices.¹⁶ The magnitude of the melting point shift is consistent with DSC curves of confined [BMIM][PF₆] within ionogels and capillary confinement within mesoporous silica.^{34,69} It has been widely reported that the shift of melting or cold crystallization peaks is an indicator that the IL has successfully been confined within a porous structure.^{17,19,34,84} This shift can increase or decrease the melting or crystallization based on many factors, some of which are the pore diameter, pore wall chemistry, synthesis method, and cation/anion components.¹⁷ Another important factor that may lead to the melting point elevation is pore wall-induced layering in the IL, which may help to stabilize crystallites in the pores.

Confinement of [BMIM][PF₆] within tethered 9 nm particles led to a further melting point decrease to 6.6 °C. Molecular dynamics simulations of [BMIM][PF₆] on a quartz surface suggest that the imidazolium cation from the IL preferentially interacts with the negatively charged silica surface, forming a layer of cations.²⁰ Tethering the IL replaces the partially deprotonated Si–O–H bonds on the pore walls with a positively charged imidazolium ring. The smaller pore diameter and increased solid–liquid surface free energy after tethering would cause the melting point to decrease further, according to the Gibbs–Thomson equation (eq 1). The water contact angle is a good indicator of interfacial tension where the increase in water contact angle corresponds with an increase in surface energy. A glass plate has a water contact

angle of 12° when cleaned with chromic acid,⁸⁵ whereas the contact angle is 24° for a glass surface functionalized with a monolayer of [TMS-MIM][Cl],²⁹ signaling that the surface has become more hydrophobic. Density functional theory and AFM measurements of adhesion force by Zhang et al. also show that for [BMIM][PF₆], a cationic surface has a weaker interaction with the IL than neutral or anionic surfaces, which corresponds to a higher IL contact angle (and increased surface energy).⁸⁶ The combined effect of increasing surface energy and smaller pore diameter reduces the melting point of [BMIM][PF₆] more than confinement in bare silica pores.

3.4. Confocal Microscopy. The diffusion of the fluorescent dyes within nanoconfined ionic liquids was investigated by using the FRAP technique. Structures of the IL and dye molecules are shown in Figure 5. The particles were immersed in water immediately prior to the measurements to maintain a refractive index close to that of the oil objective. This was necessary for the microscope to photobleach the correct area. Without water, the particles were unable to be photobleached due to significant differences in refractive index between oil and air. The equator or vertical axis midpoint of the particle was used to accurately measure the diffusivity, as illustrated in Figure 1. The values of diffusivity were found by first determining the effective radius of the Gaussian bleach profile in the first collected image (Figure 1c) and fitting the fluorescence intensity vs time data using eq 3, as illustrated in Figure 1d and discussed in the Methods section. The resulting diffusivity values are summarized in Figure 6. The large standard deviation in diffusivity is related to the heterogeneity of the confined IL/dye system. Local heating from the laser could potentially contribute to heterogeneity, but because of the low concentration of fluorescent dye in the sample, temperature changes during FRAP can generally be neglected.⁸⁷ The low concentration of dye would mitigate significant increase in temperature, and the small area would mean that temperature increases would dissipate rapidly into the surrounding areas; therefore, we do not anticipate laser heating to be a concern. Ionic liquids have been reported to

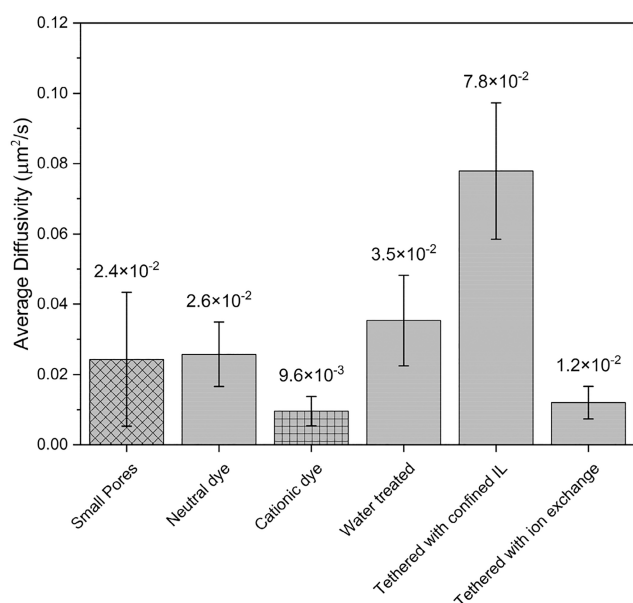


Figure 6. Diffusivities of DCM and R6G within ionic liquid confined in microparticles measured by the FRAP method in confocal microscopy as a function of pore diameter, dye charge, water effects, pore wall tethering, and tethering with ion exchange. Error bars represent the standard deviation of each data set. The default dye used was DCM, and the default pore diameter was 9 nm. In the small pore case (leftmost bar with a diamond fill pattern), the dye used is DCM and the pore diameter is 5.4 nm. In the cationic dye case (third from left, grid pattern), the pore diameter is 9 nm and the dye used is R6G.

have a microheterogeneous nature which may contribute to the uncertainty.^{41,87,88} Additionally, mesoporous silica particles can have heterogeneity between different individual particle FRAP measurements.⁸⁹ The diffusion coefficients reported here are an ensemble average over the area probed, which after bleaching gives an effective radius of several μm (e.g., Figure 1c).⁹⁰

The effect of confinement and pore size on the diffusivity of the neutral DCM probe was measured by using mesoporous silica particles with pore diameters of 5.4 and 9 nm. The resulting DCM diffusivities are 0.024 ± 0.019 and $0.026 \pm 0.0091 \mu\text{m}^2/\text{s}$ for 5.4 and 9 nm pore diameters, respectively. It was hypothesized that a reduction in pore diameter would result in more IL/pore wall interactions due to the increased surface area and that therefore the dye would not be able to diffuse as easily through the system. The lack of a statistically significant difference between small and large pores could mean that the relative amount of IL interacting with the pore wall does not play a large role in the diffusivity of the neutral probe moving through the nanopore. As the size of the pore is decreased from 9 to 5.4 nm, the surface area increases from 446 to 654 m^2/g , and the SAXS pore-to-pore distances are 11.1 and 8 nm for the particles, respectively. This implies that the pore wall thickness is similar between the two samples (approximately 2.1 and 2.6 nm pore walls for 9 and 5.4 nm diameters, respectively) and the smaller pores are more constricted, but it has little effect on diffusivity. Because both particles have pores large enough to permit dye molecules to pass each other in the pores, a more extreme change in the conditions may be required to significantly alter diffusivity in the confined system.

Compared to its diffusivity in bulk [BMIM][PF₆], the DCM dye diffuses nearly 2 orders of magnitude slower in nanoconfined ILs. Werner et al. studied the bulk diffusion of DCM in [BMIM][PF₆] using FCS and reported a value of $1.74 \mu\text{m}^2/\text{s}$.⁴³ Similarly, Patra and Semanta fit FCS data using two-component model and reported a diffusion coefficient of $1.30 \pm 0.4 \mu\text{m}^2/\text{s}$.⁹¹ It is worth mentioning that the absolute diffusion coefficients can vary between FCS and FRAP measurements of the same system because the techniques measure diffusion in different time windows, length scales, and contexts.⁴⁴ The diffusivity of 4-(*N,N'*-dimethylamino)-benzonitrile (DMABN) in bulk [BMIM][PF₆] measured by FRAP was reported as $0.25 \mu\text{m}^2/\text{s}$ by Santhosh et al.⁴¹ The DMABN probe is less bulky than DCM, but the 5-fold difference in diffusivity highlights the potential discrepancies between the two techniques. A comparison to the self-diffusion of the IL is also useful. NMR has shown that the room temperature bulk-phase self-diffusion coefficients of the cation and anion are 5.2 and $3.9 \mu\text{m}^2/\text{s}$, respectively.⁹² The larger size of the dye molecule is predicted by the Stokes–Einstein (SE) equation to give slower tracer diffusion compared to self-diffusion.⁹³ Further, a survey of tracer diffusion in bulk ionic liquids has shown that small molecules exhibit systematically faster diffusion than the SE equation predicts, approaching the SE model as solute size increases.^{93,94} As with the reduction of dye diffusivity, the self-diffusion coefficient of the cation and anions is also reduced when confined. In one study, NMR showed that the self-diffusion coefficient of 1-hexyl-3-methylimidazolium PF₆ was slowed by an order of magnitude when confined into mesoporous silica with 7.5 nm diameter pores due to hydrogen bonding of the IL to the pore walls.⁹⁵ Importantly, the comparison of self-diffusion to probe diffusion reveals the unique approach and novelty of using FRAP to quantify diffusivity within confined ILs. Other techniques such as NMR probe the IL self-diffusion coefficient within the IL while FRAP probes diffusion of a solute within the IL. The comparison between FRAP of the DMABN, DCM, and other dyes to the self-diffusion probed by NMR demonstrates that the diffusion of a solute can be distinctly different than that of the IL, which necessitates using techniques to probe solute transport independently.

There are two possible reasons for the reduction of the diffusivity of the dye when the IL is confined. The first is the increase in the path length that a molecule needs to travel in a porous material due to the pore volume and tortuosity. Using reconstructed electron tomography models, Hochstrasser et al.⁹⁶ developed an empirical model of these effects for SBA-15. Using the Stokes radius of DCM (0.44 nm)⁴³ and assuming the pore fraction (0.69) and tortuosity (1.41) of SBA-15 reported by Hochstrasser et al., $D_{\text{eff}}/D_{\text{bulk}}$ values of 0.24 and 0.33 are calculated for DCM diffusion in 5.4 and 9 nm diameter pores, respectively. This does not explain the ratio observed of ~ 0.013 , so molecular interactions with the pore wall should be considered as a second factor affecting diffusion of probes in silica-nanoconfined ILs. Pal et al. performed molecular dynamics simulations of [BMIM][PF₆] confined in uncharged silica nanopores and found a significant reduction in overall diffusivity (by about a factor of 3) relative to bulk IL and a spatially dependent reduction by 2 orders of magnitude in the layer of ions closest to the pore wall.⁴⁶ The layering observed by Pal et al. was driven by hydrogen bonding of PF₆[−] with silanols, and Sha et al. showed that ion layering and its effects on reducing diffusivity of [BMIM][PF₆] are increased

by adding a negative charge to the confining surface.³³ Thus, ion layering contributes to a reduction of diffusivity both by increasing the local viscosity and by reducing the accessible pore volume with layers of rigid ions near the pore wall. Consistent with this idea, it has been reported that confinement in even relatively large conical pores (without tortuosity) reduced the diffusivity of fluorescein isothiocyanate in a 20% glycerol/water mixture by over 10-fold.⁹⁷

With a lack of effect of pore size on diffusivity observed for the 5.4 and 9 nm pores, the remaining confined IL systems are compared to that of the neutrally charged probe DCM in [BMIM][PF₆] loaded in silica particles with 9 nm pores. The effect of the charge of the dye was examined for the cationic dye (R6G) in microparticles with the same pore size. A comparison of the dye structures can be found in Figure 5. The effect of hydration of the neutral probe molecule was studied by immersing the particles in water approximately 24 h prior to imaging; this sample was prepared to study the robustness of the system to an aqueous environment. An IL-like molecule ([TMS-MIM][Cl]) was covalently attached to the pore wall in the tethered-9 nm silica sample, changing the pore wall chemistry and interactions between the IL and pore wall. Lastly, to see the effect of the counterion, the [TMS-MIM][Cl] group was first tethered, and then the [Cl][−] anion was exchanged for [PF₆][−] to match the anion with the confined IL.

The diffusivities for the neutral (DCM) and cationic (R6G) dyes were found to be 0.026 ± 0.0091 and 0.011 ± 0.0049 $\mu\text{m}^2/\text{s}$, respectively. Werner et al. found that cationic dye diffusion was $\approx 20\%$ slower than neutral in bulk [BMIM][PF₆] using FCS,⁴³ and our results show a larger reduction in diffusivity (almost 60%) in the confined system.⁹¹ The difference between the dyes in the bulk phase may be from interactions between the cationic dye and anion of the IL but could also be from a size difference because the cationic dye is bulkier than the neutral dye.⁴² However, in the confined system, the effect of tortuosity due to the larger size of the cationic dye would be expected to decrease the diffusivity by only about 10%,⁹⁶ so that is still not enough to explain the observed decrease. The two-component FCS diffusion model of rhodamine 123 and DCM in [BMIM][PF₆] shows that there are different contributions to diffusivity from dye in the alkyl tail or ionic domains formed within the IL, corresponding to contributions of a fast and slow component, respectively.⁹¹ Patra et al. found that the association of the R123 molecule with the IL through hydrogen bonding with the IL causes the dye to reside primarily in the polar region of the IL and thus possess a greater contribution from the slow diffusivity component, leading to lower diffusivity.⁹¹ Despite being similar in size, R123 has a smaller diffusion coefficient than DCM when dissolved in an IL,⁹¹ so it is more likely that IL–dye interactions cause the decrease in diffusivity. The cationic dye could interact with either the negatively charged [PF₆][−] from the IL or the silica pore walls as shown in Figure 5. The compounded effect of these interactions, along with possible effects of ion layering induced by the pores (increasing local viscosity at the pore walls), likely leads to a slower diffusivity in the confined system as there are more potential sites for association with the dye.

The particles were immersed in water for confocal microscopy to provide an appropriate refractive index medium for imaging. Prior to immersion, the amount of water dissolved within the confined IL is not expected to be significant because

the particles are dried with heat (80 °C) and under vacuum. While [BMIM][PF₆] is hydrophobic, there could potentially be an effect of a small amount of water taken up by the IL. Not only is the presence of water a potential factor in our diffusivity measurements due to the addition of water for refractive index matching, but understanding the effect of humidity and/or aqueous environments on the diffusivity in IL's is important to its commercial application as a confined solvent. During a typical experiment, the mixture of water with the particles for refractive index matching was not hypothesized to occur because [BMIM][PF₆] is hydrophobic and does not adsorb a significant amount of water over the short time scale of the experiment (<2 h).⁹⁸ To address the potential for water to be adsorbed during the measurements, the confined ILs were immersed in water with no agitation for 24 h to study the response of the system to water to allow the mixture to reach equilibrium. The maximum solubility of water into [BMIM][PF₆] is 2.67% (w/w) at room temperature,⁹⁹ and water in the vapor phase reaches equilibrium with liquid phase within 24 h.¹⁰⁰ After this treatment, the diffusivity was 0.035 ± 0.013 $\mu\text{m}^2/\text{s}$, which is not statistically different from the value measured for the sample that was not saturated with water (0.026 ± 0.0091 $\mu\text{m}^2/\text{s}$). The increase in diffusivity indicates that the water uptake is slow, but it is likely that water entered the pores over the time of exposure (≈ 24 h) and reached saturation levels with [BMIM][PF₆].^{99,100} Like prior studies, the average increase in diffusivity is most likely from the decrease in viscosity of the mixture.⁴³ The apparent increase in diffusivity after water exposure in our system is consistent with the diffusivity increase by FCS reported by Werner et al. upon adding 3 vol % water to bulk phase [BMIM][PF₆] (1.74 to 3.0 $\mu\text{m}^2/\text{s}$).⁴³ Schröder used voltammetry to show that the diffusivity of a neutral redox probe molecule *N,N,N',N'*-tetramethyl-*p*-phenylenediamine in bulk phase [BMIM][PF₆] doubled from 2.6 to 5.2 $\mu\text{m}^2/\text{s}$ when the concentration of water was raised from 0 to 5% (w/w).¹⁰¹ They also report an increase of an order of magnitude from 1.1 to 10 $\mu\text{m}^2/\text{s}$ when using the cationic probe methyl viologen and 6% (w/w) of water in the IL,¹⁰¹ although the drastic increase in diffusivity may be unreliable because the concentration of water is much higher than its solubility limit in [BMIM][PF₆].⁴³ In our experiments, the IL was not mixed with water before confinement, so the modest increase observed may be limited by the equilibrium solubility of water.

When the 9 nm pore walls were tethered with an IL-like molecule, [TMS-MIM][Cl], the diffusivity of the DCM dye in the confined IL was found to increase to 0.078 ± 0.019 $\mu\text{m}^2/\text{s}$. The significant diffusivity increase is likely a result of a change in molecular interactions caused by charge and hydrophobicity changes within the pores because smaller pores did not present a significant change. When [TMS-MIM][Cl] is covalently attached to the pore wall, the negative charge of the silica surface is screened or reversed by the imidazolium groups, and the environment within the pore becomes more hydrophobic (similar to the bulk IL). The surface interaction of [BMIM][PF₆] has been shown to become weaker as the silica surface becomes less negative due to functionalization with cationic groups.¹⁰² Because DCM is a hydrophobic dye, it should be more mobile in this environment. The result is consistent with prior observations of the transport of 1,1'-dioctadecyl-4,4'-bipyridinium dibromide (DBD), a hydrophobic redox probe with a high octanol/water partition coefficient, meaning it is more lipophilic than hydrophilic.⁶² DBD did not encounter

significant hindrance to diffusion after the pore walls of a silica thin film with 8 nm pores were tethered with [TMS-MIM][Cl] while a more hydrophilic redox probe experienced diffusion resistance through the same pores.⁶² The DBD probe is larger and matches the charge of the pore wall, while DCM is smaller and is a neutral dye. This result is also supported by recent AFM measurements of surface interactions between ILs and functionalized silica.¹⁰² Interaction forces of the IL with the bare silica surface are much stronger than with a positively charged surface with a quaternary ammonium group.¹⁰² It would follow that the lower interaction force between the IL and silica would lead to a larger diffusivity. Tethering is often used to ensure that the IL does not leach from the system during use, so this counterintuitive result is significant because it shows that a reduction in pore diameter due to tethering does not necessarily lead to a decrease in diffusivity in a condensed pore-confined phase.

As the last system variable studied, the counterion of the tethering group was exchanged so that a mixture of chloride and hexafluorophosphate groups would not be present in the pores. The tethered particles underwent ion exchange to change the [Cl][−] ion to [PF₆][−] by exposing the functionalized particles to a NaPF₆ solution. While this was hypothesized to further accentuate the hydrophobic environment of the nanopores compared to the tethered-9 nm system with confined IL, it resulted in a significant decrease in effective diffusivity to $0.012 \pm 0.0046 \mu\text{m}^2/\text{s}$. As discussed above, the water contact angle for a Si/SiO₂ surface functionalized with [TMS-MIM][Cl] changed from 24° to 42° after the Cl[−] anion was exchanged with PF₆[−] on silica thin films, indicating increasing hydrophobicity that is correlated to a decrease in surface energy.^{29,86} The tethered-9 nm with a confined IL system establishes that higher hydrophobicity within the pores relative to bare silica or a positively charged surface leads to higher diffusivity, so the lowered diffusivity is unexpected. The decrease is likely related to an increase in the difficulty in loading the IL after ion exchange. In Figure S3, an image of an example particle demonstrates higher heterogeneity in dye loading compared to the other particle treatments. The fluorescence intensity (and thus dye concentration) across the particle is heterogeneous in the ion exchanged particles in contrast to the other confined IL treatments, which have a uniform intensity over the whole particle. Such heterogeneity could be caused by incomplete ion exchange, partial blocking of the pores, or damage to the pore structure during the ion exchange procedure. The bright spots outside of the particle could be evidence of pore structure modification where some fused particles have sintered during processing, making parts of the particles inaccessible. Whatever the cause of the heterogeneous loading, it causes an increase in the effective tortuosity for a network of pores incompletely filled with IL.

4. CONCLUSIONS

Confocal fluorescence microscopy, differential scanning calorimetry, FTIR, and nitrogen porosimetry were used to study the pore structure and its relation to the diffusivity of probe dye molecules in [BMIM][PF₆] confined within mesoporous silica microparticles. Nitrogen porosimetry was used to determine the pore diameter and texture of the silica microparticles and show a difference in pore diameters due to hydrothermal treatment and a decrease in pore diameter after tethering the pores with methylimidazolium chloride groups. DSC showed that the IL was successfully confined within the

pores of the silica. Shifts in the melting and cold crystallization temperatures to lower temperature upon confinement were observed, consistent with what is expected from the Gibbs–Thomson equation. FTIR also demonstrated that ILs were successfully confined, as indicated by small shifts in the bands associated with IL due to attraction to the pore wall.

The diffusivity of the neutral fluorescent probe DCM was first measured in 9 nm pore diameter particles to show the ability of the FRAP technique to effectively measure the diffusivity. The diffusivity of the DCM in the confined system is about 2 orders of magnitude lower than in the bulk phase, which is typical of what is observed for diffusion of probe molecules in confined fluids. The similarity of DCM diffusivity between small and large pores demonstrates that the system is robust to pore diameter changes within relatively large mesopores used for this comparison. The cationic dye (R6G) had a diffusivity lower than that of the neutral DCM because of the interactions of the dye with the IL and pore wall. A simulated water immersion experiment did not lead to significant changes in diffusivity, but the average diffusivity was somewhat higher after increasing the water content of the system by immersion for 24 h. Water may be absorbed to a small extent by the IL within the pores to reduce the effective viscosity of the confined IL system, leading to a small reduction of viscosity and an increase in diffusivity.

The most significant effects on the diffusivity of the DCM probe are found when an IL-like methylimidazolium group is tethered to the pore surface. When the surface is functionalized, the increase in diffusivity is likely caused by the increase in hydrophobicity within the pores and weaker interactions with the pore walls. Because DCM is a hydrophobic probe, it moves more easily in this environment. Exchanging the anion of the tethered species to match the IL anion led to a drastic decrease in diffusion and a highly heterogeneous dye distribution in the particles. The ion exchange may have not been successful for all tethered sites and led to the observed heterogeneity in dye concentration, or the pore structure may have been altered during the ion exchange procedure.

Understanding the transport of dye molecules in confined ILs is crucial to the development of supported IL systems in separations, heterogeneous catalysis, and batteries. Bulk ILs have faster transport properties, but the challenges of using them due to their high viscosity and cost make confinement a desirable solution. This study complements the findings published using fluorescence methods to study diffusion in bulk systems. The large decrease in diffusivity of confined IL compared to that of the bulk phase highlights the importance of pore wall interactions, charge, and electrostatics when designing confined systems. However, this decrease in diffusivity does not detract from the advantage provided by the short diffusion path of ILs confined to nanoporous materials. For example, we can consider use of [BMIM][PF₆] supported in 9 nm pores of a mesoporous silica membrane for perstraction of pollutants such as bis-phenol A. Based on the reported distribution coefficient from water into [BMIM][PF₆] of $K_D = 400^5$ and assuming a 1 μm thick active layer ($=t$) with the diffusivity measured for the tethered, confined IL, the permeance would be $P = K_D D/t = 400 \times (0.0779 \mu\text{m}^2/\text{s})/(1 \mu\text{m}) = 31.2 \mu\text{m}/\text{s}$. This compares favorably to polymer-based supported ionic liquid membranes, where permeance values have been reported to be on the order of 10^{-2} – $10^{-3} \mu\text{m}/\text{s}$.¹⁰³

Water and pore size did not significantly alter the diffusion in confinement, meaning these design parameters are less

important in confined IL systems than in bulk ILs, expanding their possible applications. For example, CO₂ capture in flue gases would be an appropriate application due to the high affinity of many ILs to CO₂. Because the pore size does not significantly limit diffusion, the particle synthesis for this application is not as stringent, and water in flue gas would not significantly change the diffusive properties of CO₂ adsorption, leading to longer material life span. Tethering also shows that the transport properties can be modified by adjusting the support/IL interfacial chemistry to achieve slower or faster transport. From this work, it is clear that confined ILs offer high flexibility in their design parameters which makes them more viable to use in various applications, such as perstraction of organic solutes or CO₂ capture.

The FRAP technique in confocal microscopy was demonstrated to be a useful technique to study solvent transport within confined ILs within nanopores. The extension of this technique to confined ILs is significant. While existing studies show that FRAP can be used in bulk phase ILs, this work demonstrates that the significant reduction in diffusivity due to confinement can be measured and characterized by the technique. Importantly, the solute diffusion is measured rather than the IL cation and anion self-diffusion, which are distinctly different and must be understood to further understand ILs as a solvent. FRAP could be extended to other confined systems such as with deep eutectic solvents (DESs).¹⁰⁴ Often considered analogous to ILs, DESs also typically suffer from high viscosity and benefit from confinement. DES systems are becoming increasingly studied in scientific literature, and it is imperative that their properties be well understood.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.4c01018>.

Full list of calculated diffusivity values and the coefficients of determination for each sample in confocal microscopy of the confined dye/IL combination (Table S1); reduced SAXS patterns and one 2D pattern for the support materials (Figure S1); full FTIR spectra from Figure 3 (Figure S2); representative confocal images of particles used in this study for each treatment (Figure S3) (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the United States National Science Foundation (NSF) under Grant CBET-1604491 for the initial development of tethering and ionic liquid confinement methods in mesoporous silica. Measurement and analysis of diffusion of fluorescent probes in the microparticles were financially supported by Honeywell Corporation. Additional analysis based on literature models was performed under NSF Grant OIA-2218054. Part of this work was performed in part at the University of Kentucky Electron Microscopy Center which belongs to the National Science Foundation NNCI Kentucky Multiscale Manufacturing and Nano Integration Node, supported by ECCS-1542174.

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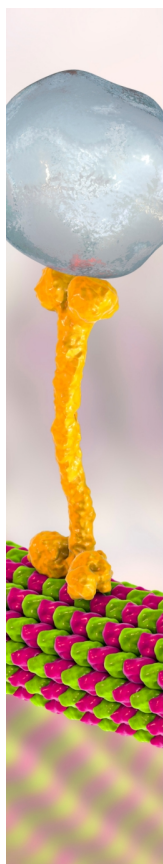
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